

Student ID: #

Islamic University of Technology (IUT)
Department of Computer Science and Engineering (CSE)
Third Quiz Examination, Winter Semester 2024-2025
Course Title: Geometry and Differential Calculus
Course No.: MATH 4141, Section No.: SWE

Full Marks: 20

Exam Time: 30 Minutes

*N.B.: Answer all **FOUR** questions. Numbers given in the right side indicate full marks.*

1. Check whether the equation $x^2 - 5xy + 4y^2 + x + 2y - 2 = 0$ represents a pair of straight line or not. If so, find the equation of the lines, their point of intersection, angles between the lines, the equation of the bisectors of the angles between them, and finally sketch the lines showing the necessary information. [7]
2. Find the projection of the joining of two points $(-1, -1, 3)$ and $(2, 0, 1)$ on the line through the points $(-7, 5, 3)$ and $(2, 6, 8)$. [3]
3. Find the perpendicular distance of the point $(-2, 3, 4)$ from the straight line through the point $(-1, 3, 2)$ whose direction ratios are $12, 3, -4$. [4]
4. Show that the triangle formed by the lines $ax^2 + 2hxy + by^2 = 0$ and $lx + my = 1$ is right angled if $(a+b)(al^2 + 2hlm + bm^2) = 0$. [6]

The End